

NAMIBIA SENIOR SECONDARY CERTIFICATE

PHYSICS ORDINARY LEVEL

6118/1

PAPER 1 Multiple Choice

45 minutes

Marks 40

2024

Additional Materials: Multiple choice answer sheet
Non-programmable calculator
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write in soft pencil.
- Make sure that you receive the multiple choice answer sheet with **your examination number** on it.
- There are **forty** questions on this paper. Answer **all** questions.
- For each question, there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.
- If you want to change an answer, thoroughly erase the one you wish to delete.
- **Read the instructions on the answer sheet carefully.**
- Each correct answer will score one mark.
- Any rough working should be done in this booklet.
- All questions in this paper carry equal marks.
- You may use a non-programmable calculator.

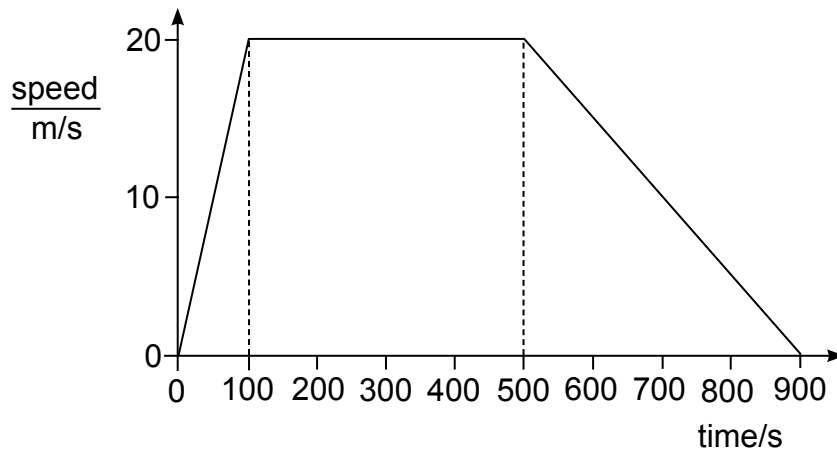
This document consists of **12** printed pages.



Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 Which instrument is used to measure accurately the inner diameter of a wedding ring?
- A 30 cm ruler
 - B metre rule
 - C micrometer screw gauge
 - D vernier calipers
- 2 An object is released from rest and falls to Earth. During its fall, the object is affected by air resistance. The air resistance eventually reaches a constant value. Which description about successive stages of the motion of the object is correct?
- A constant acceleration, then constant deceleration
 - B constant deceleration, then zero acceleration
 - C decreasing acceleration, then constant deceleration
 - D decreasing acceleration, then zero acceleration
- 3 The graph represents the motion of a train travelling between two stations.



Which statement about the train is correct?

- A Its acceleration reached 4000 m/s^2 in the first 100 s.
 - B Its deceleration occurred over a 4000 m distance.
 - C It travelled 2000 m in the first 100 s.
 - D It travelled 10 000 m at constant speed.
- 4 Which quantity has the same unit as force?
- A electromotive force
 - B moment of force
 - C weight of an object
 - D work done on an object

5 Which words describe why a physical quantity is classified as a vector quantity?

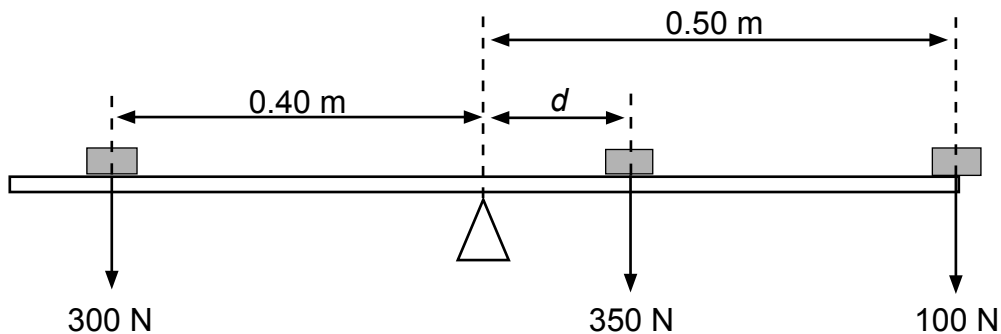
- A direction only
- B size but no direction
- C both size and direction
- D neither size nor direction

6 Pressure is force acting per unit area.

Which changes produce the same pressure?

- A double the area and double the force
- B double the area and halve the force
- C double the area and make the force four times bigger
- D halve the area and double the force

7 A uniform beam is pivoted at its centre. The beam is balanced by three weights in the positions shown.

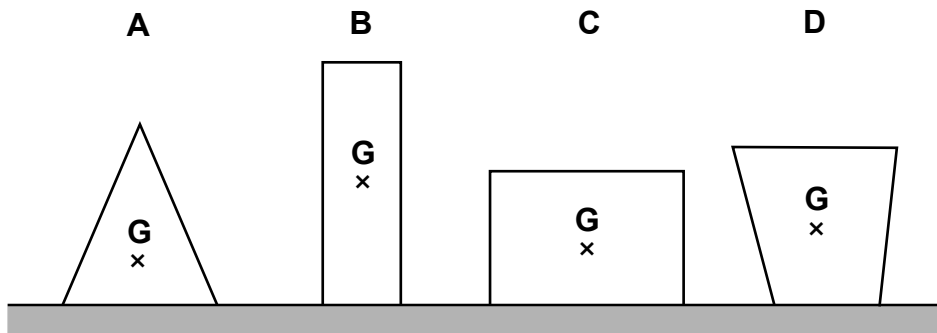


What is the length d ?

- A 0.020 m
- B 0.050 m
- C 0.20 m
- D 0.48 m

8 Four objects of equal mass rest on a smooth surface. The centre of mass of each object is labelled **G**.

Which object is the least stable?



- 9 An object of 400 kg mass is moving at a speed v m/s. It collides head-on with another object of 200 kg mass moving at a speed of 100 m/s in the opposite direction. The two objects stick together after collision, moving at a speed of 80 m/s. What was the speed, v m/s, of the 400 kg object before collision if momentum has been conserved?

A 50 m/s
 B 70 m/s
 C 80 m/s
 D 100 m/s

- 10 The soccer ball of mass 0.5 kg was dropped from a height of 3.0 m above the ground. Air resistance is negligible.

Which row correctly gives the maximum kinetic energy attained by the ball?

	amount in joules	position
A	7.5	as it lands
B	7.5	at the highest height
C	15	as it lands
D	15	at the highest height

- 11 Which type of energy is described as energy possessed due to position above the ground?

A chemical energy
 B kinetic energy
 C potential energy
 D strain energy

- 12 The diagram shows the only three forces acting on an object.



What is the resultant force on the object?

A 10 N towards the right
 B 20 N towards the left
 C 30 N towards the left
 D 40 N towards the right

- 13 Which statement defines specific latent heat?

A The heat absorbed to change the state of 1 kg of a substance.
 B The heat absorbed to change the state of a substance.
 C The heat required to raise the temperature of 1 kg of a substance by 1 K.
 D The heat required to raise the temperature of a substance by 1 K.

- 14** To mark a scale on a thermometer, standard temperatures known as fixed points are needed.

Which of the following provides a fixed point?

- A** the room temperature
- B** the temperature inside a freezer
- C** the temperature of pure melting ice
- D** the temperature of warm water

- 15** Each of the two statements below describes a change of state due to cooling of the substance.

change 1: Molecules stop moving throughout the substance and just vibrate about fixed positions.

change 2: Molecules move closer together but continue to travel throughout the substance.

Which changes of state do these statements describe?

	change 1	change 2
A	condensation	melting
B	condensation	solidification
C	solidification	condensation
D	solidification	melting

- 16** At a certain temperature, all particles stop moving.

Which statement describes this temperature?

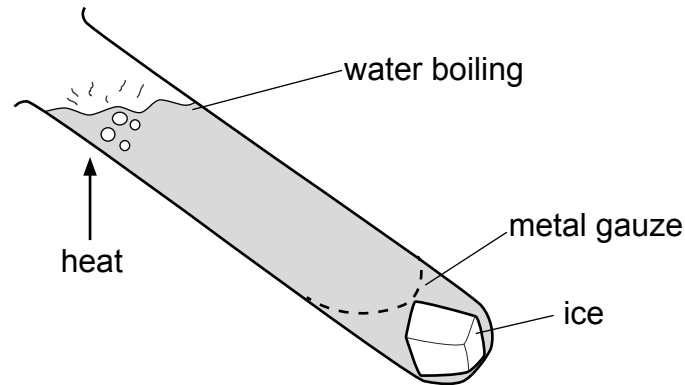
- A** The absolute zero and it is 0 degree Celsius.
- B** The absolute zero and it is 0 Kelvin.
- C** The freezing point and it is 0 degree Celsius.
- D** The freezing point and it is 0 Kelvin.

- 17** An object made of cast iron has a mass of 2.0 kg. 23 000 J of energy was used to increase the temperature of this object from 50 °C to 75 °C.

What is the specific heat capacity of cast iron?

- A** 153 J / (kg °C)
- B** 230 J / (kg °C)
- C** 460 J / (kg °C)
- D** 920 J / (kg °C)

- 18 Ice is trapped by a piece of metal gauze at the bottom of a tube containing water. The water at the top of the tube boils before the ice at the bottom of the tube melts.



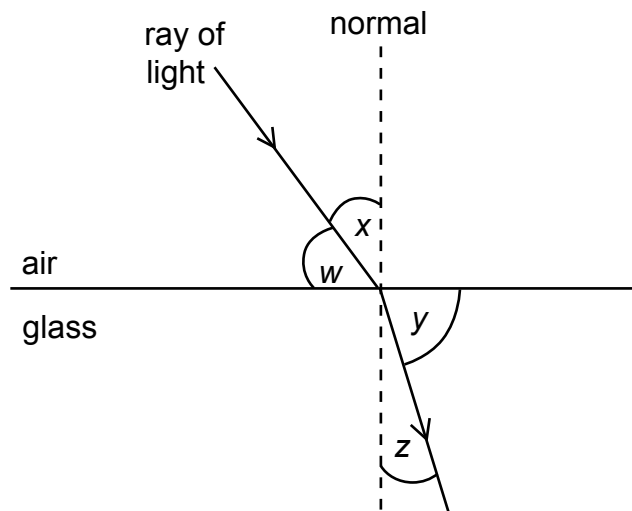
Why does the melting of the ice occur slowly?

- A Hot water is denser than cold water.
 - B Ice is a poor absorber of heat.
 - C Metal gauze does not allow heat to pass through.
 - D Water is a poor conductor of heat.
- 19 What is the name given to the transfer of heat energy which **does not** require a medium?
- A conduction
 - B convection
 - C evaporation
 - D radiation
- 20 Which row shows an example of a transverse wave and an example of a longitudinal wave?

	transverse	longitudinal
A	light	radio
B	light	sound
C	sound	surface of water
D	surface of water	light

- 21 Which electromagnetic waves have the shortest wavelength?
- A gamma rays
 - B infra-red waves
 - C microwaves
 - D radio waves

- 22** An ultraviolet wave has a wavelength of 3.0×10^{-7} m and a frequency of 1.0×10^{15} Hz.
What is the speed of this wave?
- A** 3.3×10^{-23} m/s
B 3.0×10^8 m/s
C 3.0×10^9 m/s
D 3.3×10^{22} m/s
- 23** In which substance does sound travel fastest?
- A** air
B steel
C vacuum
D water
- 24** The diagram shows a ray of light refracting as it travels from air into glass.

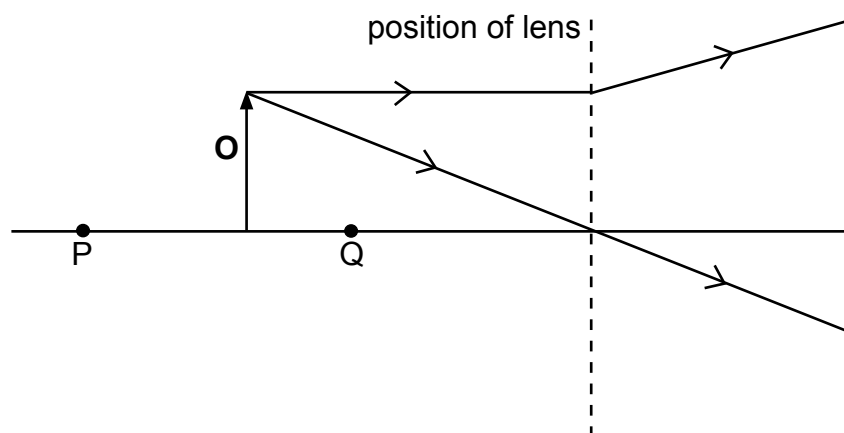


Which angle represents the angle of refraction?

- A** w
B x
C y
D z

- 25** An object **O** is placed in front of a lens.

The ray diagram shows the paths of two light rays that start at the tip of the object.



Which point P or Q is a principal focus and what type of image is formed?

	principal focus	type of image formed
A	point P	real
B	point P	virtual
C	point Q	real
D	point Q	virtual

- 26** What is the approximate range of audible sound frequencies for a healthy human ear?

- A** 20 Hz to 20 000 Hz only
- B** 200 Hz to 2 000 Hz only
- C** 200 Hz to 20 000 Hz only
- D** 2 000 Hz to 200 000 Hz only

- 27** A pulse of sound is produced by a bat in the direction of a mountain.

The sound travels through the air and is reflected by a vertical surface on the side of the mountain. The sound travels to this vertical side of the mountain and back in 6.0 s. The mountain is 1000 m away from the bat.

What is the speed of this sound?

- A** 167 m/s
- B** 333 m/s
- C** 3 000 m/s
- D** 6 000 m/s

- 28** What is the unit for the energy used by an electrical appliance?

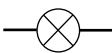
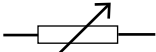
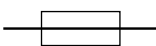
- A** coulombs per second
- B** joules per second
- C** kilowatt-hour
- D** volt per ampere

- 29 A metal conductor is connected to a battery.

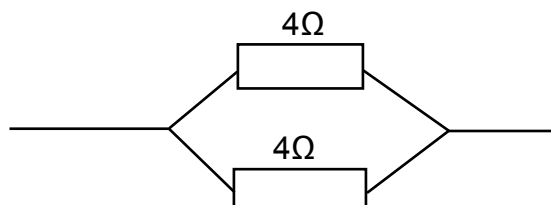
Which statement describes the current in the metal conductor?

- A It is a flow of electrons from the negative to the positive terminal.
- B It is a flow of electrons from the positive to the negative terminal.
- C It is a flow of protons from the negative to the positive terminal.
- D It is a flow of protons from the positive to the negative terminal.

- 30 Which electrical symbol represents a variable resistor?

- A 
- B 
- C 
- D 

- 31 The diagram shows part of a circuit consisting of two identical resistors. The circuit is supplied by a 4 V source.



What is the total current in the circuit?

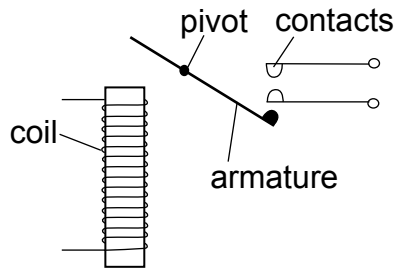
- A 0.5 A
 - B 1.0 A
 - C 2.0 A
 - D 4.0 A
- 32 The table describes four different resistance wires. They are all made from the same metal.
- Which wire has the smallest resistance?

	length of wire/m	diameter of wire/mm
A	3.0	1.0
B	3.0	1.3
C	4.0	1.0
D	4.0	1.3

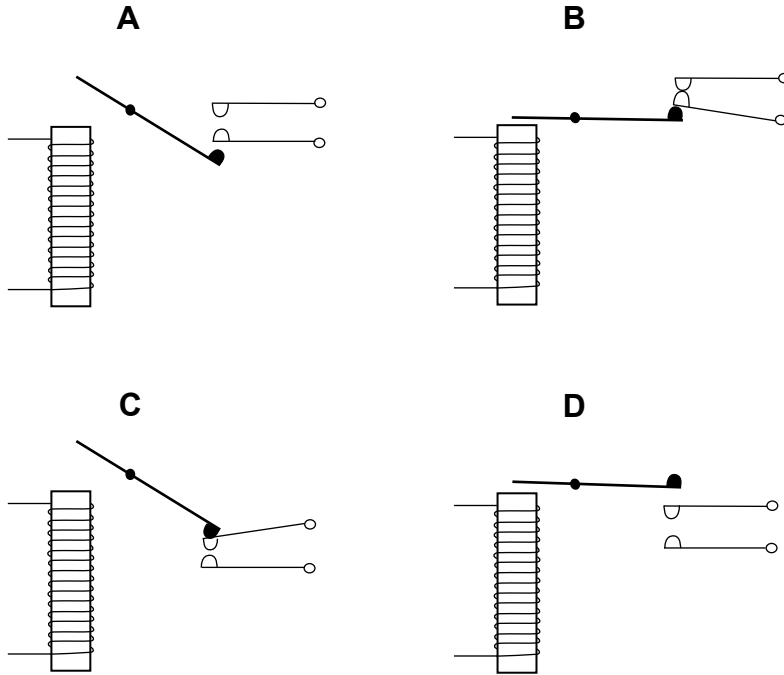
- 33 Which colour correctly identifies the neutral wire in a mains plug?

- A blue
- B brown
- C green
- D red

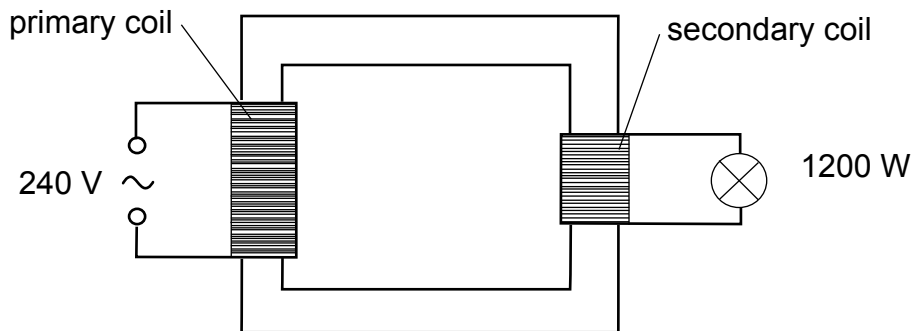
- 34** The diagram shows the arrangement of the coil, armature and contacts of a relay.



Which diagram shows this arrangement when there is current flowing in the coil?



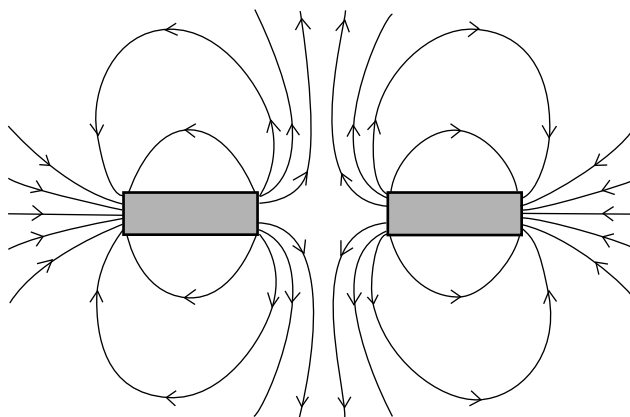
- 35** A 100% efficient step-down transformer is used to light a lamp from a 240 V mains supply. The transformer has a power output of 1200 watts.



What is the current in the primary coil?

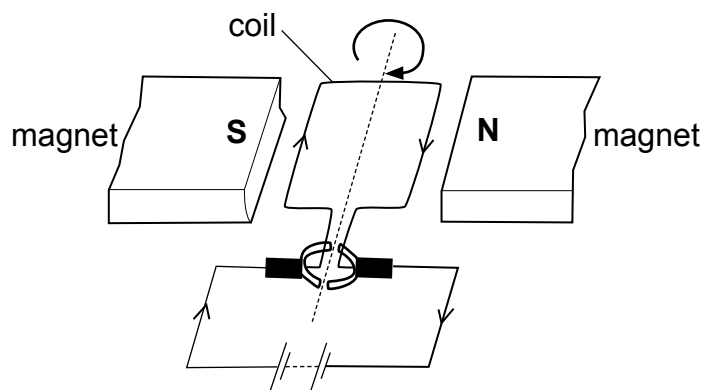
- A** 0.2 A
- B** 0.4 A
- C** 2.5 A
- D** 5.0 A

- 36** The diagram shows the magnetic field around two bar magnets.



Which statement about the two magnets is correct?

- A** The magnets attract because the poles facing each other are both north.
 - B** The magnets attract because the poles facing each other are north and south.
 - C** The magnets repel because the poles facing each other are both north.
 - D** The magnets repel because the poles facing each other are north and south.
- 37** What is the effect of using a split-ring commutator?
- A** it ensures that the current is the same in all parts of a series circuit
 - B** it generates an alternating electric current
 - C** it produces a force on a current-carrying coil
 - D** it reverses the direction of the current in the coil of a motor
- 38** The diagram shows a simple d.c. electric motor which is rotating.



Which change makes the motor rotate faster?

- A** increasing the number of turns on the coil
- B** removing the magnets
- C** reversing the battery
- D** reversing the polarity of the magnets

- 39 How many nucleons are in one neutral atom of Thorium isotope ${}^{234}_{90}\text{Th}$?
- A 90
B 142
C 144
D 234

- 40 A radioactive source emitted a particle.

Which row indicates that the particle emitted is an alpha?

	ionising effect	deflection in an electric field
A	strong	from positive to negative
B	strong	no deflection
C	weak	from positive to negative
D	weak	no deflection

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